

Haodi (Woody) Hu

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[Personal Website](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

Summary

Ph.D. in embodied AI and learning-based control, specializing in reinforcement learning, VLM/VLA post-training, and World-Action Models (WAM) for long-horizon robotic manipulation and tactile-grounded dexterous manipulation, with publications in CoRL, ICRA, T-RO, and RA-L.

Core Strengths

Research Areas	World-action models, VLM/VLA post-training, reinforcement learning, tactile dexterous manipulation, learning-based control, embodied decision-making
Systems / Scale	RL optimization, tactile manipulation, robotics dataset development, real-world robot deployment, cloud computing
Tools	PyTorch, ROS2, Isaac Sim, Cosmos3, GR00T, OpenPI / $\pi_{0.5}$, PPO, RGB-D perception, CUDA, LoRA, Linux
Programming	Python, C++, MATLAB, SQL

Experience

Senior Machine Learning Engineer <i>Sandisk</i>	Jun 2026 – Present <i>San Jose, CA</i>
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- Lead R&D on reinforcement-learning-based optimization and post-training of vision-language models for domain-specific applications.
- Develop RL systems for supply-chain and semiconductor-production optimization, formulating planning and production decisions as long-horizon sequential control problems under operational constraints and uncertainty.
- Post-train VLMs for a highly domain-specific Japanese–English technical translator, grounding specialized manufacturing terminology in linguistic and visual context.

Research Scientist Intern <i>Mitsubishi Electric Research Laboratories (MERL)</i>	Jan 2026 – Jun 2026 <i>Boston, MA</i>
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- Developed ReCoVLA, a failure-conditioned residual RL framework that freezes a fine-tuned $\pi_{0.5}$ -family base policy and learns corrective actions from VLA latent features.
- Designed a Qwen3-VL-guided reward compiler that converts failure descriptors into stage-gated rewards, improving simulation success from 36.7% to 66.7% and Q-score from 0.56 to 0.83.
- Achieved 61.7% success and 0.75 Q-score in zero-shot physical Fetch experiments across sorting, disposal, and contact-rich toolbox recovery.
- Developed TacSushi, coupling a Cosmos3-style world-action model with tactile risk prediction and receding-horizon control for dexterous manipulation of deformable food using a four-finger Shadow Hand.

Machine Learning Engineer Intern <i>Sandisk</i>	Jun 2025 – Dec 2025 <i>San Jose, CA</i>
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- Designed and deployed PPO-based RL systems for large-scale sequential decision-making under uncertainty in supply chain optimization, improving performance by 24% over prior baselines.
- Built scalable RL and LLM training/inference pipelines, reducing latency by 41% and improving factual grounding by 13% with retrieval-augmented generation.
- Built an RL-based inventory control system with estimated business impact exceeding \$20M.

Research Highlights

ReCoVLA: VLM-Guided Reward Compilation for VLA Recovery

MERL

Embodied AI, Mobile Manipulation

- Built a residual VLA recovery system combining frozen $\pi_{0.5}$ -style execution, Qwen3-VL failure interpretation, and PPO-style residual correction in the VLA latent space.
- Constructed a semantic failure catalog and reward compiler that grounds VLM descriptors into object-state reward components with stage-aware gates.
- Evaluated ReCoVLA on vegetable sorting, soda-can disposal, and contact-rich toolbox organization, achieving the strongest simulation and physical Fetch results among tested methods.

TacSushi: Dexterous Hand Sushi Making with World-Action Model

MERL

World Models, Dexterous Manipulation

- Fine-tuned a Cosmos3-style WAM to generate 16-DoF Shadow Hand action chunks and predict visual consequences for rice shaping, seaweed wrapping, and sashimi topping placement.
- Trained an action-conditioned tactile network to predict contact maintenance, contact risk, over-compression, and uncertainty; fused tactile safety with visual progress in risk-constrained MPC.
- Built a dexterous manipulation dataset of 323 successful demonstrations and 50 failed teleoperation trials; achieved 68% average real-robot success on in-distribution tasks and 38% zero-shot success on unseen ingredient variants.

Real-World Robotic Loco-manipulation on Challeneger Terrain

USC RoboLAND

Loco-manipulation, Deformable Terrains

- Built autonomous data-collection and RGB-D perception pipelines with gantry-based and quadrupedal robot platforms for terrain-mediated manipulation.
- Trained ViT-based models to predict granular media avalanche responses from legged excavation actions and RGB-D terrain observations.
- Developed diffusion-based granular dynamics predictors and control policies for quadrupedal robots to reposition rocks through strategic sand avalanches and improve locomotion on granular slopes.

Publications

- **Haodi Hu**, Kaen Kogashi, Toshiaki Koike-Akino. “TacSushi: Tactile-Grounded World-Action Modeling for Dexterous Sushi Manipulation.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2027 (submitted).
- **Haodi Hu**, Chung-Ta Huang, Jing Liu, Ye Wang, Kei Suzuki, Matthew Brand, Toshiaki Koike-Akino. “ReCoVLA: VLM-Guided Reward Compilation for Failure Recovery in Vision-Language-Action Policies.” *Conference on Robot Learning (CoRL)*, 2026 (submitted).
- **Haodi Hu**, Yue Wu, Daniel Seita, Feifei Qian. “Granular Loco-manipulation: Repositioning Rocks Through Strategic Sand Avalanche.” *Conference on Robot Learning (CoRL)*, 2025.
- **Haodi Hu**, Xingjue Liao, Wuhao Du, Feifei Qian. “Multi-robot Connection Towards Collective Obstacle Field Traversal.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- **Haodi Hu**, Feifei Qian, Daniel Seita. “Learning Granular Media Avalanche Behavior for Indirectly Manipulating Obstacles on a Granular Slope.” *Conference on Robot Learning (CoRL)*, 2024.
- **Haodi Hu**, Feifei Qian. “Obstacle-Aided Trajectory Control of a Quadrupedal Robot Through Sequential Gait Composition.” *IEEE Transactions on Robotics (T-RO)*, 2024.
- Kaustav Chakraborty, **Haodi Hu**, Matthew Kvalheim, Feifei Qian. “Planning of Obstacle-Aided Navigation for Multi-Legged Robots Using a Sampling-Based Method Over Directed Graphs.” *IEEE Robotics and Automation Letters (RA-L)*, 2022.

Education

University of Southern California

Ph.D. in Electrical Engineering

May 2026

Los Angeles, CA

- Research focus: embodied AI, VLA systems, world-action models, reinforcement learning, tactile dexterous manipulation, and mobile manipulation.
- Thesis title: Towards Obstacle-Aided Locomotion: Robot Mobility and Terrain Manipulation Through Leg-Ground Interactions.

University of Southern California

M.S. in Electrical Engineering

May 2021

Los Angeles, CA

Northeast Forest University

B.S. in Electrical Engineering

Jun 2019

Harbin, China

Honors, Teaching & Service

- IEEE Access Exceptional Reviewer Recognition, 2026; USC Viterbi Ph.D. Student Fellowship Award, 2021; USC Viterbi CURVE Mentor Award, 2022–2025; NEFU Excellent Graduate Award, 2019.
- Co-organizer, 5th Workshop on Representations and Manipulating Deformable Objects, ICRA 2025.
- Teaching Assistant for Robot Mobility (EE599), Deep Learning (EE541), and MOS VLSI Circuit Design (EE477L), 2022–2025.
- Mentored USC CURVE and student researchers on robotics projects, research preparation, and conference presentation.